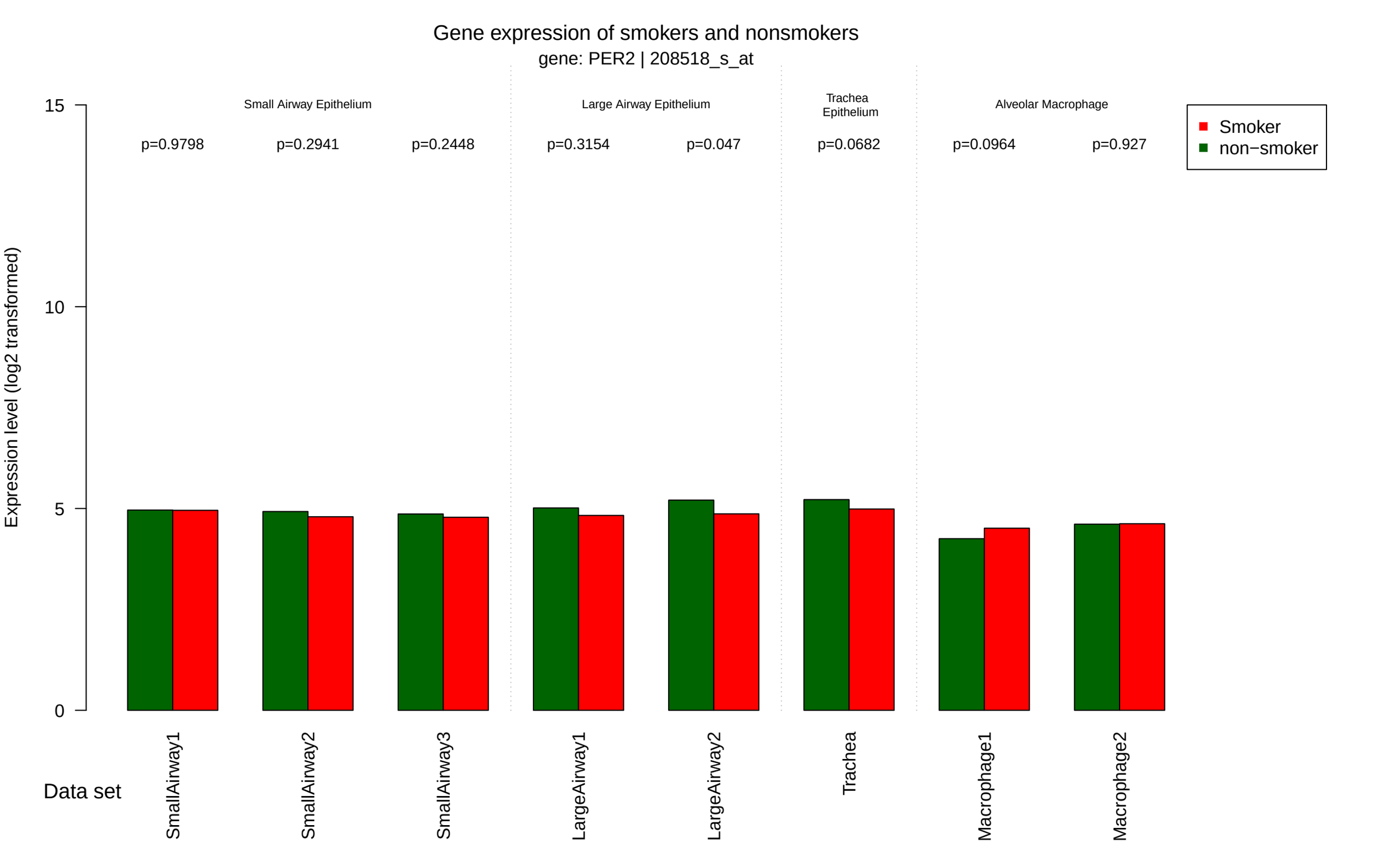
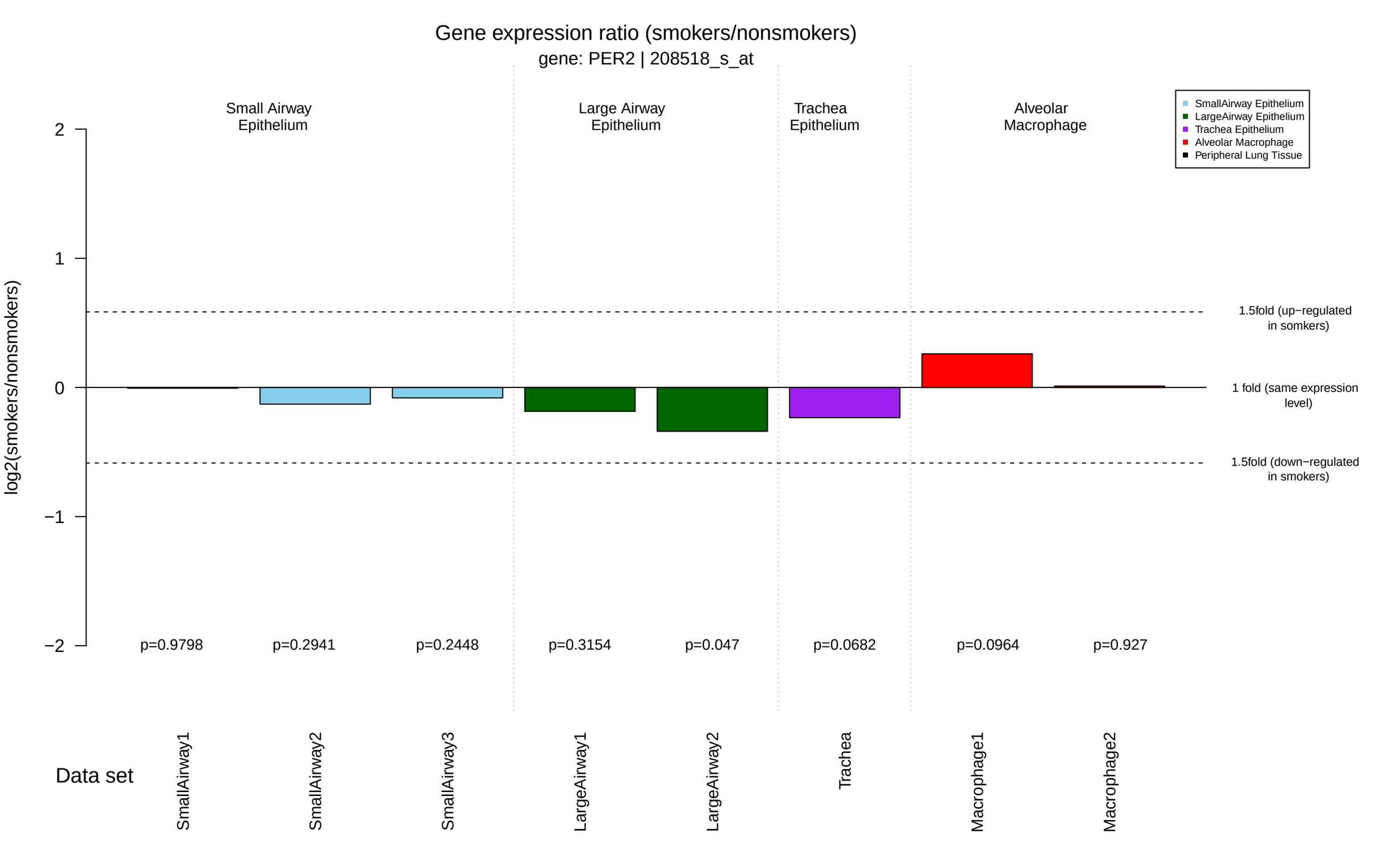
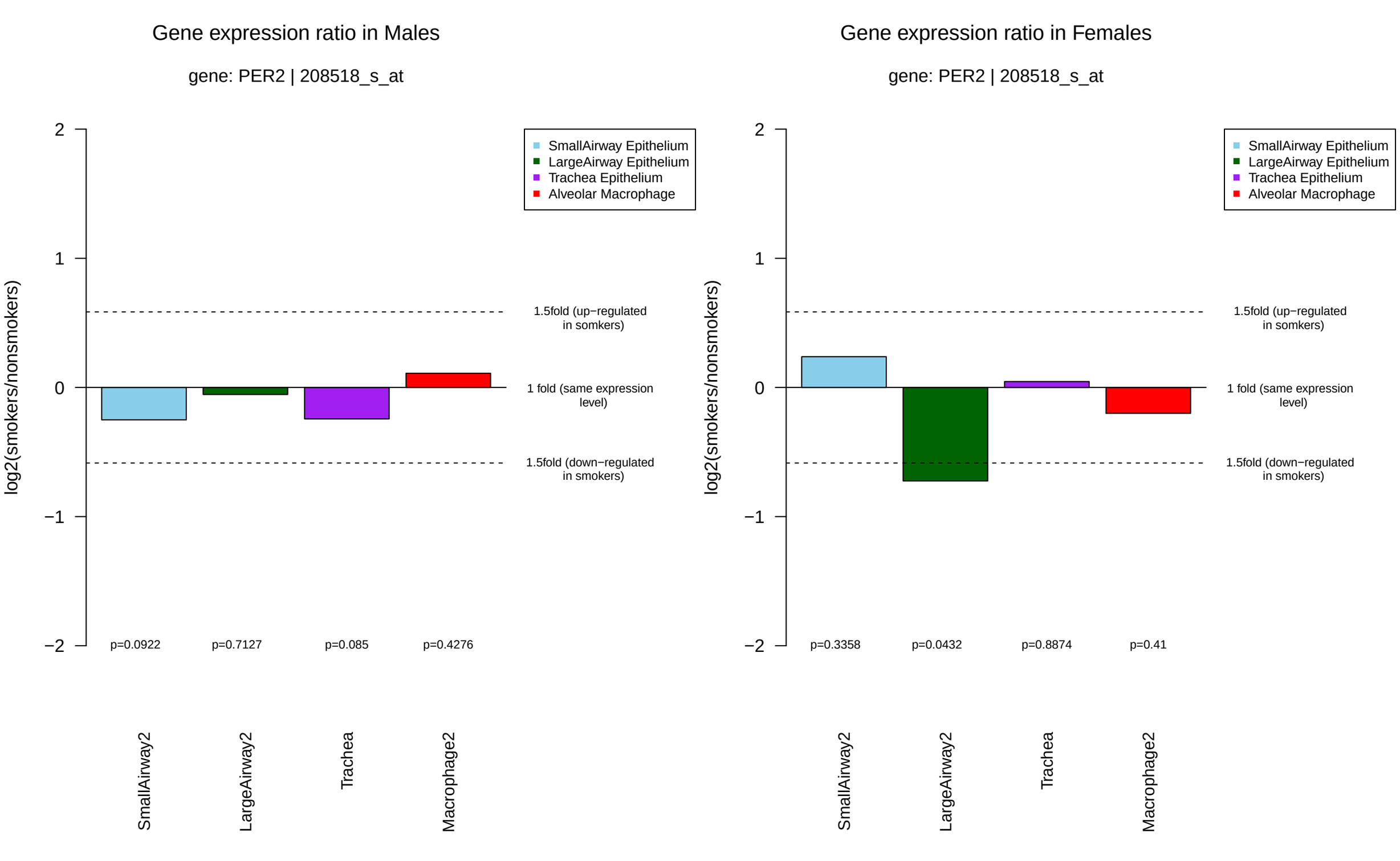
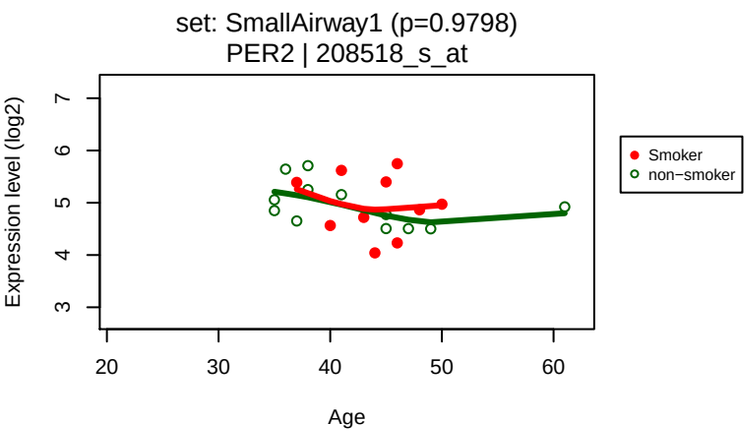
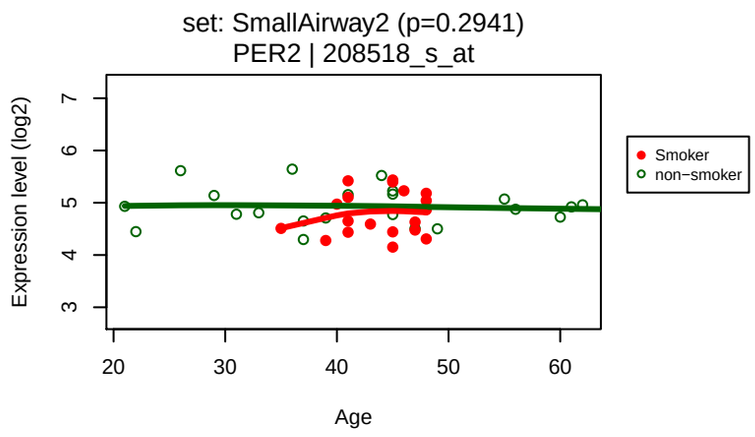
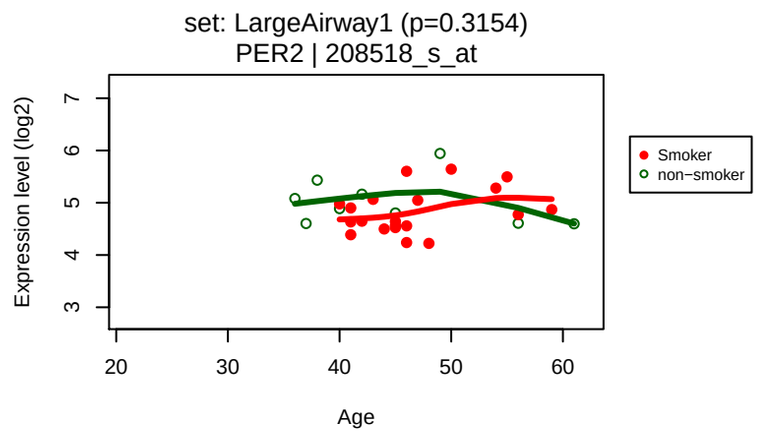

Gene expression of smokers and nonsmokers
gene: PER2 | 208518_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9798
p=0.2941
p=0.2448
p=0.3154
p=0.047
p=0.0682
p=0.0964
p=0.927
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

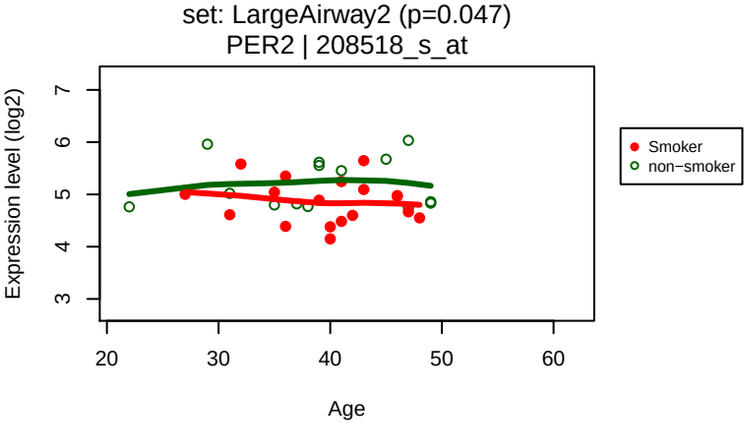

Gene expression ratio (smokers/nonsmokers)
gene: PER2 | 208518_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9798
p=0.2941
p=0.2448
p=0.3154
p=0.047
p=0.0682
p=0.0964
p=0.927
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

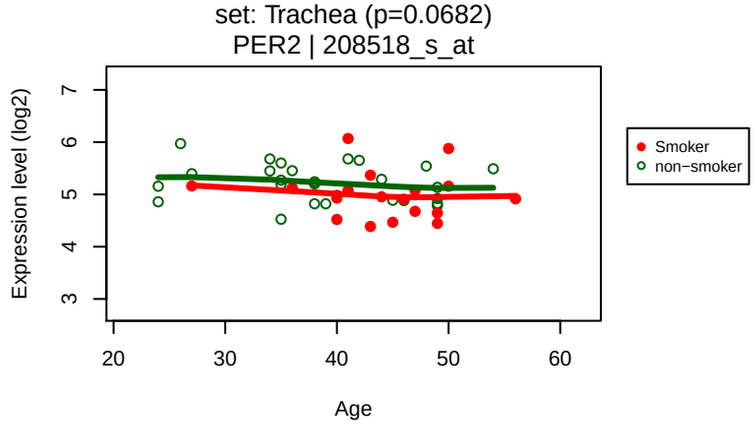

Gene expression ratio in Males
gene: PER2 | 208518_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0922
p=0.7127
p=0.085
p=0.4276
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PER2 | 208518_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3358
p=0.0432
p=0.8874
p=0.41
SmallAirway2
LargeAirway2
Trachea
Macrophage2

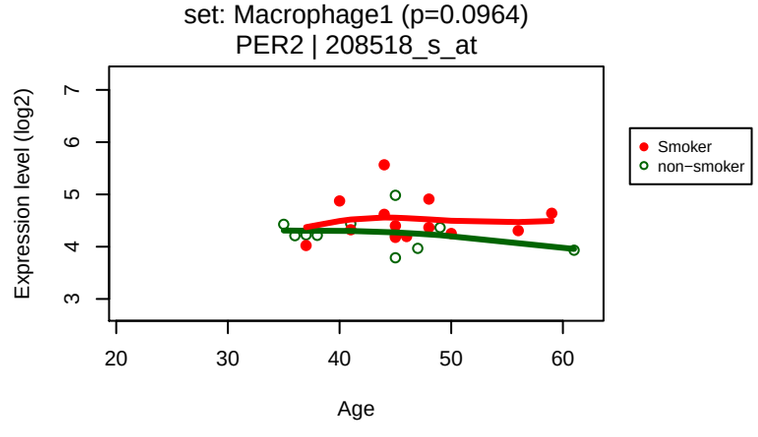

set: SmallAirway1 (p=0.9798)
PER2 | 208518_s_at
Expression level (log2)
Age
Smoker
non−smoker

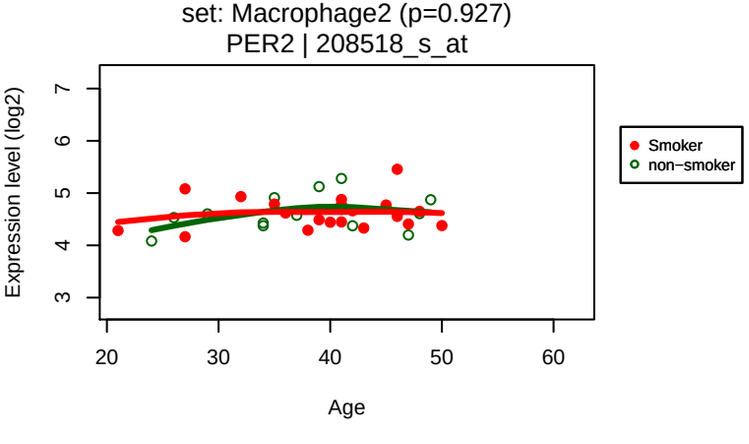

set: SmallAirway2 (p=0.2941)
PER2 | 208518_s_at
Expression level (log2)
Age
Smoker
non−smoker

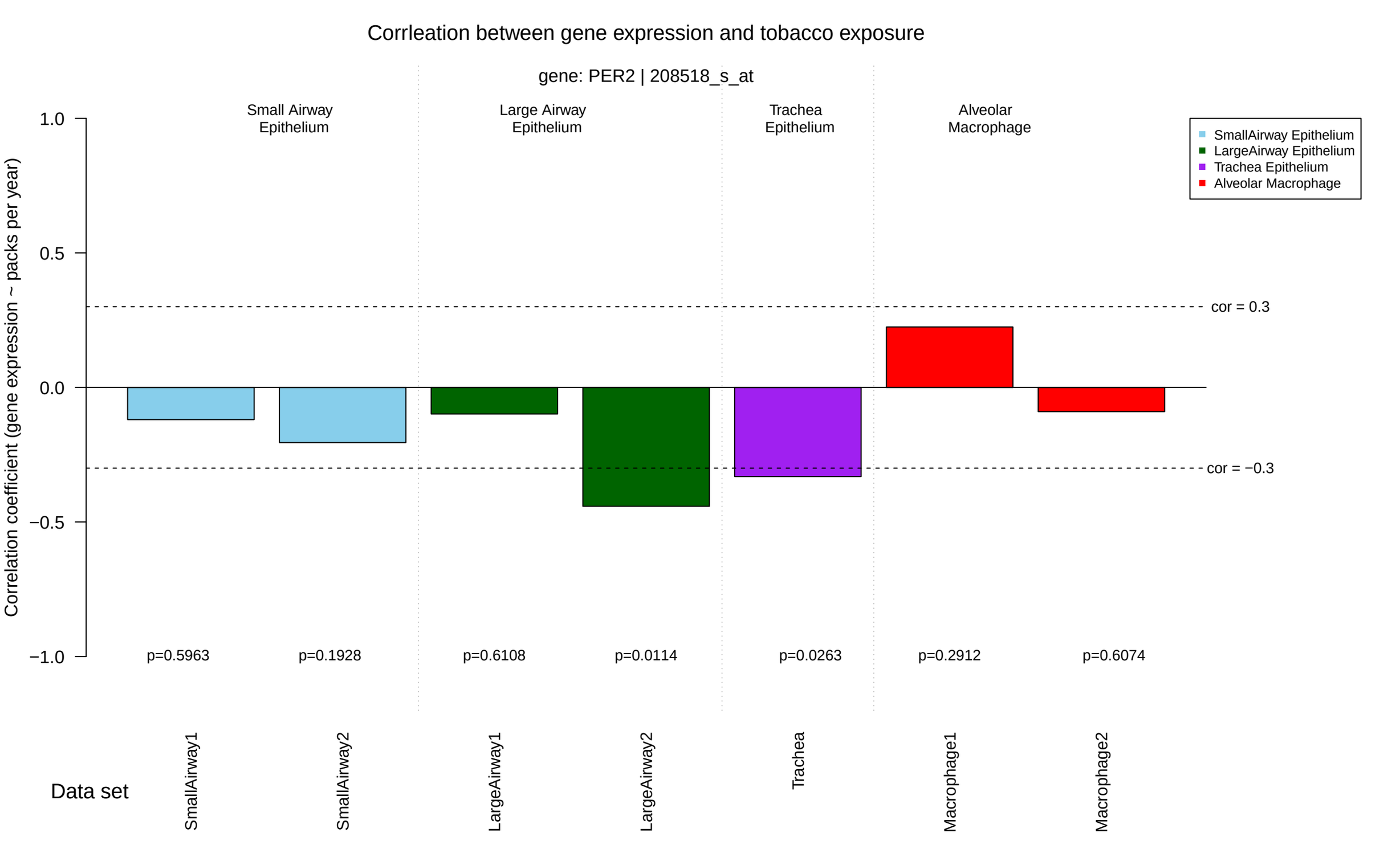

set: LargeAirway1 (p=0.3154)
PER2 | 208518_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.047)
PER2 | 208518_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0682)
PER2 | 208518_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0964)
PER2 | 208518_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.927)
PER2 | 208518_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PER2 | 208518_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5963
p=0.1928
p=0.6108
p=0.0114
p=0.0263
p=0.2912
p=0.6074
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2